The first finite-copy correction in two-parameter qubit estimation

Tingyu Zhang[1, *] and Yan Cen[1, †]

[1] *Department of Physics, Fudan University, Shanghai, China*

Joint measurements on finitely many quantum-state copies must balance information about incompatible spin components. Their approach to infinite-copy precision depends on the constraints of a complete measurement. We determine the first finite-copy correction to the attainable locally unbiased risk for two-parameter qubit estimation at the maximally mixed state. Optimizing over joint measurements on N copies, the minimum total mean squared error multiplied by N is $2 + \sqrt{8/\pi}\,N^{-1/2} + o(N^{-1/2})$. The central result is that the common convex roof of the two transverse spin variances at the maximally mixed spin-j state $I/(2j+1)$ equals $j/2 + o(j)$. We construct Gaussian columns with position-dependent widths whose populations integrate to one exactly. A finite phase average then produces a complete positive operator-valued measure for every finite spin. Uniform energy estimates and a matching sum-of-squares lower bound establish the optimal correction. An explicit width sequence and a score estimator attain it while preserving exact local unbiasedness for every finite N. The strict finite-copy Nagaoka–Hayashi gap for $N \geq 10$ therefore contributes only $o(N^{-1/2})$ to the normalized risk.

I. INTRODUCTION

Finite-copy precision in multiparameter quantum estimation depends on how information about incompatible observables can be collected in a single measurement. Even when the infinite-copy limit is known, constructing a complete measurement that attains its leading correction remains a separate problem. We address this question for the two Cartesian components of a qubit at the maximally mixed state.

Conic formulations characterize the attainable local risk and allow reconstruction of complete measurements [1]. Schur–Weyl reduction further simplifies the finite-copy optimization [2]. These methods provide a finite-dimensional framework, but convergence of a measurement grid at fixed N does not determine how the optimal risk approaches its infinite-copy limit. Here we seek both the correction as the number of copies grows and an explicit family of positive operator-valued measures (POVMs) attaining it.

Planar spin squeezing and Gaussian descriptions of low-variance spin states provide the local ingredients [3, 4]. In particular, the fixed-squeezing asymptotics of Ref. [4, Theorem 4], after rotation and optimization, give the local cost $j|x|$ at a fixed nonzero latitude x. Its formal average is $j\int_{-1}^{1}|x|\,dx/2 = j/2$. Realizing this average requires an exact decomposition of I/d: low-variance states chosen separately need not have the required average populations. We enforce this constraint by normalizing each Gaussian column before averaging its phase, and then control the resulting variance across the entire spin sector.

This construction, together with a lower bound valid for all POVMs, gives the optimal normalized risk $C_N = 2 + \sqrt{8/\pi}\,N^{-1/2} + o(N^{-1/2})$. Every finite member of the explicit family is complete and admits an exactly locally unbiased estimator. The same analysis shows that the strict finite-copy Nagaoka–Hayashi (NH) gap for $N \geq 10$ is subleading on the $N^{-1/2}$ scale. The proof combines exact population normalization, a uniform energy estimate, and averaging over the Schur sectors.

II. MODEL AND ATTAINABLE PRECISION

Consider the full-rank local qubit model

$$\rho_{x,y} = \frac{1}{2}(I + x\sigma_x + y\sigma_y), \qquad x^2 + y^2 < 1, \tag{1}$$

at $(x, y) = (0, 0)$, where σ_x and σ_y are Pauli matrices. The parameters are Cartesian components of the Bloch vector. For each positive integer N, a single arbitrary joint POVM is applied to $\rho_{x,y}^{\otimes N}$. Its outcome may be discrete or continuous. We require the estimator $\widehat{\theta} = (\widehat{x}, \widehat{y})$ to satisfy

$$\mathbb{E}_0\widehat{\theta}_a = 0, \qquad \partial_{\theta_b}\mathbb{E}_\theta\widehat{\theta}_a\Big|_0 = \delta_{ab}, \qquad \theta = (x, y), \tag{2}$$

and define

$$C_N = N \inf_{M,\widehat{\theta}} \mathbb{E}_0(\widehat{x}^2 + \widehat{y}^2). \tag{3}$$

Both the measurement and estimator may depend on the reference point. There is no restriction on ancillary systems, classical randomization, the number of outcomes, or the rank of the effects.

Write J_x, J_y, J_z for the spin-j angular momentum operators, in units with $\hbar = 1$, and let $d = 2j + 1$. The relevant common variance convex roof is

$$V_j = \min_{\sum_a p_a|\psi_a\rangle\langle\psi_a| = I/d} \sum_a p_a\big(\mathrm{Var}_{\psi_a} J_x + \mathrm{Var}_{\psi_a} J_y\big). \tag{4}$$

Here the p_a are probabilities and the ψ_a are normalized pure states. Both variances use the same decomposition. Such common convex roofs are part of the general

* 23300190021@m.fudan.edu.cn

† Corresponding author: cenyan@fudan.edu.cn

framework of uncertainty relations based on pure state decompositions [5].

Let $\mathcal{J}_N = \{j_{\min}, j_{\min}+1, \ldots, N/2\}$, with $j_{\min} = 0$ for even N and $1/2$ for odd N. The Schur decomposition and its probabilities at the reference point are

$$\begin{aligned}(\mathbb{C}^2)^{\otimes N} &= \bigoplus_{j\in\mathcal{J}_N} \mathcal{H}_j \otimes \mathbb{C}^{\nu_j},\\ \nu_j &= \binom{N}{N/2-j} - \binom{N}{N/2-j-1},\\ P_{N,j} &= \frac{(2j+1)\nu_j}{2^N}.\end{aligned} \tag{5}$$

An out-of-range binomial coefficient is zero. Conditional on j, the reference state and its two tangent operators are

$$Q_j = \frac{I}{d}, \qquad \dot{Q}_{j,x} = \frac{2J_x}{d}, \qquad \dot{Q}_{j,y} = \frac{2J_y}{d}. \tag{6}$$

The probabilities $P_{N,j}$ have zero first derivatives in both parameters. Write $\mathbb{E}_N$ for expectation under $P_{N,j}$.

Lemma 1. *If T_j is the maximum classical Fisher trace in the spin-j model (6), then*

$$\begin{aligned}T_N &= \sum_j P_{N,j} T_j, \qquad C_N = \frac{4N}{T_N},\\ T_j &= \frac{8}{3} j(j+1) - 4V_j.\end{aligned} \tag{7}$$

These identities hold for the full class of POVMs, with no additional sample-size limit.

The proof, including the treatment of arbitrary complex and continuous-outcome POVMs, is given in Appendix A. The maximum is attained, so the optimization defines an attainable local risk for each finite N.

III. OPTIMAL FINITE-COPY CORRECTION

Theorem 2. *As $j \to \infty$ through integer or half-integer spins,*

$$V_j = \frac{j}{2} + o(j). \tag{8}$$

For the model (1), optimized over all joint POVMs and all estimators satisfying (2),

$$\boxed{C_N = 2 + \sqrt{\frac{8}{\pi}}\, N^{-1/2} + o(N^{-1/2}).} \tag{9}$$

The Gaussian-column measurement in Section IV, with

$$\epsilon_j = \frac{1}{\log(e + j + 1/2)}, \qquad j > 0, \tag{10}$$

and the estimator in (19) are well-defined, complete, and exactly locally unbiased for every positive integer N. Their risk has the same expansion (9).

The coefficient in (9) is fixed by the cost of a complete identity decomposition within each spin sector. Appendix B gives a sum-of-squares certificate for all POVMs, with lower bound $L_j = j/2 + o(j)$. Appendix C proves that the exactly complete Gaussian family has average variance $W_j^{\mathrm{G}} = j/2 + o(j)$. Thus $L_j \le V_j \le W_j^{\mathrm{G}}$ determines the spin asymptotic.

The Schur mean spin satisfies $\mathbb{E}_N j/\sqrt{N} \to \sqrt{2/\pi}$. For any fixed spin sequence $r_j = o(j)$, Appendix D proves $\mathbb{E}_N r_j = o(\sqrt{N})$ by a direct finite-head and large-spin split. The variance cost, linear in j to leading order, therefore becomes a Fisher-information loss of order $\sqrt{N}$. Equation (7) converts this loss into the $N^{-1/2}$ correction to the normalized risk. In particular, the proof covers both parities of N and includes all small-spin sectors.

IV. COMPLETE MEASUREMENT

Fix $j > 0$ and set $J = j + 1/2$. Initially let $b \in C^2([-1,1])$ be positive, and define

$$\begin{aligned}a(x) &= (1-x^2)b(x), \qquad S(s) = Ja(s/J),\\ v_s(t) &= (2\pi S(s))^{-1/4} \exp\left[-\frac{(t-s)^2}{4S(s)}\right],\\ &\qquad |s| \le j.\end{aligned} \tag{11}$$

Use the discrete columns $v_m(t)$, $m = -j, \ldots, j$. With $c_m = \sqrt{(j-m)(j+m+1)}$, put

$$\begin{aligned}s_j(t) &= \sum_m v_m(t)^2,\\ h_j(t) &= \sum_{m=-j}^{j-1} c_m v_m(t) v_{m+1}(t).\end{aligned} \tag{12}$$

For an integer $L \ge \max(d, 3)$ set $\phi_\ell = 2\pi\ell/L$, $\ell = 0, \ldots, L-1$, and define the POVM density

$$\begin{aligned}E_j(dt, \ell) &= \frac{1}{L} |v_{t,\ell}\rangle\langle v_{t,\ell}|\, dt,\\ |v_{t,\ell}\rangle &= \sum_m v_m(t) e^{-im\phi_\ell} |m\rangle.\end{aligned} \tag{13}$$

Each $S(m) > 0$, since $|m| \le J - 1/2$. Every effect is positive, and $\int_{\mathbb{R}} v_m(t)^2\, dt = 1$. Averaging the phases eliminates all off-diagonal entries exactly; consequently

$$\sum_{\ell=0}^{L-1} \int_{\mathbb{R}} E_j(dt, \ell) = I. \tag{14}$$

Thus column normalization fixes the diagonal entries of the identity, while the phase average removes the off-diagonal entries. This completeness relation holds over the full real outcome space. The associated decomposition of I/d has density $s_j(t)/(dL)$ and normalized vectors $v_{t,\ell}/\sqrt{s_j(t)}$.

Let $W_j[b]$ be its average planar variance and put $A_m = j(j+1) - m^2$. Phase rotations preserve the sum of the two variances, so

$$W_j[b] = \frac{1}{d}\int_{\mathbb{R}} \left[\sum_m A_m v_m(t)^2 - \frac{h_j(t)^2}{s_j(t)}\right] dt. \tag{15}$$

The width is selected by balancing two contributions to the variance. Appendix C bounds (15) by a sum of nonnegative position and discrete-gradient energies, with a smaller coefficient correction. Their leading densities are $x^2 b(x)$ and $1/[4b(x)]$, respectively, for the rescaled spin coordinate $x = m/J$. Away from the equator, increasing the width raises the first contribution and lowers the second. Their balance suggests $b(x) = 1/(2|x|)$ away from the equator. We regularize it as $b_\epsilon(x) = 1/(2\sqrt{x^2+\epsilon^2})$; with (10), the measurement of Theorem 2 therefore uses the explicit width:

$$S_m = \frac{J^2 - m^2}{2\sqrt{m^2 + \epsilon_j^2 J^2}}, \qquad J = j + \frac{1}{2}, \tag{16}$$

and one may set $L = \max(2j+1, 3)$. First measure the Schur label j. For $j > 0$ apply (13), tensor the effects with the multiplicity identity, and for $j = 0$ use its identity effect. The score is the gradient of the log outcome probability at the reference point. At outcome (j, t, ℓ), its conditional value is

$$u(j,t,\ell) = \frac{2h_j(t)}{s_j(t)} \begin{pmatrix} \cos\phi_\ell \\ \sin\phi_\ell \end{pmatrix}. \tag{17}$$

Indeed $\langle v_{t,\ell}|J_+|v_{t,\ell}\rangle = e^{i\phi_\ell} h_j(t)$ with the phase convention in (13). Let $W_j^{\mathrm{G}} = W_j[b_{\epsilon_j}]$, $W_0^{\mathrm{G}} = 0$, and set

$$T_N^{\mathrm{G}} = 2N - 4\mathbb{E}_N W_j^{\mathrm{G}}. \tag{18}$$

All Gaussian columns are strictly positive, so $h_j(t) > 0$ for $j > 0$. Hence $T_N^{\mathrm{G}} > 0$ for every $N \geq 1$ and the full Fisher matrix is $(T_N^{\mathrm{G}}/2) I_2$. The estimator is

$$\widehat{\theta}(j,t,\ell) = \frac{4h_j(t)}{T_N^{\mathrm{G}} s_j(t)} \begin{pmatrix} \cos\phi_\ell \\ \sin\phi_\ell \end{pmatrix}, \qquad \widehat{\theta}(0) = 0. \tag{19}$$

It uses the total Fisher information, common to all sectors. The score argument in Lemma 1 proves exact local unbiasedness and risk $C_N^{\mathrm{G}} = 4N/T_N^{\mathrm{G}}$.

V. FINITE-COPY COMPARISON

The capped quantities in (B2) give the Nagaoka–Hayashi value for the present model:

$$C_N^{\mathrm{NH}} = \frac{N}{\sum_j P_{N,j} S_j/(2j+1)} = \frac{4N}{2N - 4\mathbb{E}_N L_j}. \tag{20}$$

For completeness, the reduction and a proof of strict separation are given in Appendix E. They imply $C_N > C_N^{\mathrm{NH}}$ for $N \geq 10$. Equations (B3) and (D4) now yield

Corollary 3. *For every $N \geq 10$, $C_N - C_N^{\mathrm{NH}} > 0$, whereas*

$$C_N - C_N^{\mathrm{NH}} = o(N^{-1/2}) \qquad (N \to \infty). \tag{21}$$

Figure 1 compares the explicit construction with the NH lower bound and the coherent-state benchmark. The left panel resolves the average variance within a spin sector; the right panel displays the resulting finite-copy risk correction. The Gaussian variance ratio at $j = 3000$ is $W_j^{\mathrm{G}}/j \simeq 0.507269852$.

The numerical values are obtained by composite trapezoidal quadrature on $[-j - 12\sqrt{\max_m S_m}, j + 12\sqrt{\max_m S_m}]$, with mesh spacing at most 0.2. This finite interval is used only for numerical integration; the measurement in (13) retains every real outcome. The largest computed column normalization residual is below 2×10^{-12}. At $j = 100$, reducing the mesh spacing to 0.1 changes W_j^{G} by approximately 4.3×10^{-14}. These checks quantify numerical consistency; the limiting constants follow from the analytic bounds.

The explicit width sequence is chosen for a uniform asymptotic guarantee and need not minimize the risk at a given finite N. For example, at $N = 400$ its risk is approximately 2.11276018, compared with the coherent-state risk 2.10872335 and the NH lower bound 2.07352470.

VI. DISCUSSION

The essential constraint is the average population of the measurement seeds. Local squeezing specifies low-variance states, whereas a valid measurement also requires their effects to sum to the identity. Column normalization imposes this condition exactly and leaves the width available to control the variance. The regularization near the equator and the estimates at the poles allow the local squeezing balance to be used throughout the spin sector. This completes the connection between the established local description [3, 4] and the attainable finite-copy correction.

The strict NH gap expresses the same completeness constraint at finite size. Seeds that saturate the capped certificate cannot resolve the identity in sufficiently large spin sectors, as shown in Appendix E. The Gaussian family satisfies completeness exactly while its variance differs from the lower bound by only $o(j)$. Schur averaging makes this an $o(\sqrt{N})$ Fisher-information deficit and an $o(N^{-1/2})$ gap in normalized risk. Thus finite-dimensional nonattainability of the NH bound is compatible with agreement at the first correction.

The logarithmic width sequence gives an explicit asymptotically optimal family, while the finite-N example in Section V shows that its convergence can be slow. Optimizing the widths for finite-copy performance and determining higher-order corrections are natural next questions. The present result concerns local estimation at the maximally mixed state; extending the construction to a

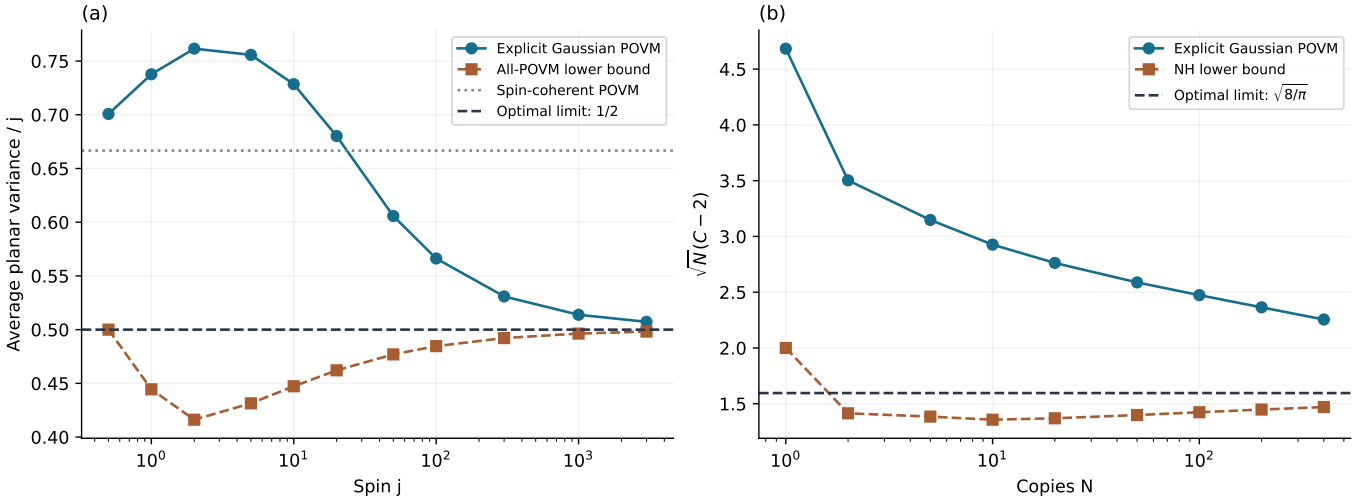


FIG. 1. Finite-copy behavior in two-parameter qubit estimation at the maximally mixed state. The Gaussian positive operator-valued measure (POVM) uses $\epsilon_j = 1/\log(e + j + 1/2)$. (a) Average planar variance divided by spin: the Gaussian value W_j^{G}/j (circles), the all-POVM lower bound L_j/j (squares), and the spin-coherent value 2/3 (dotted line). The dashed horizontal line marks the optimal limit 1/2. (b) Normalized risk correction $\sqrt{N}(C-2)$ for $C = C_N^{\mathrm{G}}$ (circles) and the Nagaoka–Hayashi (NH) lower bound $C = C_N^{\mathrm{NH}}$ (squares), with C equal to N times the total mean squared error. The dashed horizontal line marks their common limit $\sqrt{8/\pi}$. Both horizontal axes are logarithmic; lines joining numerical points guide the eye. Points are obtained by numerical quadrature, while the limiting values follow analytically. The Gaussian scheme attains the optimal first correction and is not asserted to be optimal at each finite N.

nonzero reference Bloch vector requires accounting for the corresponding change in sector states and probabilities.

VII. CONCLUSIONS

The optimal normalized risk for two-parameter qubit estimation at the maximally mixed state is $2 + \sqrt{8/\pi}\, N^{-1/2} + o(N^{-1/2})$. An exact Gaussian-column decomposition attains the underlying common planar variance $j/2 + o(j)$, and its score estimator is exactly locally unbiased for every finite N. The matching all-POVM lower bound fixes the first correction and places the strict finite-copy NH gap at a smaller asymptotic order.

DATA AND CODE AVAILABILITY

[DATA AND CODE AVAILABILITY TO BE CONFIRMED]

ACKNOWLEDGMENTS

[ACKNOWLEDGMENTS AND FUNDING TO BE COMPLETED]

DECLARATIONS

[DECLARATIONS TO BE CONFIRMED]

Appendix A: Exact statistical reduction and seed representation

This appendix proves Lemma 1 and establishes the seed representation used in the lower bound.

Proof. The global reference state and tangents are $I/2^N$ and $2J_a^{\mathrm{tot}}/2^N$. They are block diagonal in (5) and scalar on each multiplicity space. For an arbitrary effect M_z, define $E_{z|j} = \nu_j^{-1}\,\mathrm{tr}_{\mathrm{mult}}(\Pi_j M_z \Pi_j)$. Discarding the label j reproduces the original local probabilities and derivatives. Keeping it is a statistical refinement and increases Fisher information in positive semidefinite order by Cauchy–Schwarz. Conversely, any conditional POVM lifts to $E_{z|j} \otimes I_{\nu_j}$. Since $\dot{P}_{N,j} = 0$, its Fisher matrix is the weighted sum of the conditional matrices.

For any measurement, let $u(z) = \nabla \log p_\theta(z)|_0$ and $F = \mathbb{E}_0[uu^{\mathsf{T}}]$. When F is invertible, the estimator $F^{-1}u$ satisfies (2) and has covariance F^{-1}. For every locally unbiased estimator,

$$\mathbb{E}_0\big[(\widehat{\theta} - F^{-1}u)(\widehat{\theta} - F^{-1}u)^{\mathsf{T}}\big] = \mathrm{Cov}_0(\widehat{\theta}) - F^{-1} \succeq 0.$$

If F is singular, the identity $\mathbb{E}_0[\widehat{\theta}u^{\mathsf{T}}] = I_2$ is impossible. Full rank of the reference state ensures that these score identities also apply to continuous outcomes, using

the scalar trace measure as a dominating measure. In particular the scores are bounded. For positive definite F, $\operatorname{tr} F^{-1} \geq 4/\operatorname{tr} F$. Randomizing the measurement by rotations about z and retaining the rotation label gives $F = (\operatorname{tr} F/2)I_2$ without changing its trace. This proves the first two assertions.

Refining effects to rank one cannot decrease Fisher information. A pure state decomposition of I/d corresponds to the effects $E_a = dp_a|\psi_a\rangle\langle\psi_a|$. Its probability is p_a and its score is $2(\langle J_x\rangle_{\psi_a}, \langle J_y\rangle_{\psi_a})$. Completeness fixes

$$\sum_a p_a \langle J_x^2 + J_y^2\rangle_{\psi_a} = \frac{1}{d}\operatorname{tr}(J_x^2 + J_y^2) = \frac{2}{3}j(j+1).$$

Subtracting the average variance gives the last identity. □

For use in the lower bound, we give a real nonnegative seed form. In the J_z basis, indexed by $m = -j, \ldots, j$, set

$$\begin{aligned} c_m &= \sqrt{(j-m)(j+m+1)}, \qquad n_a = \sum_m v_{a,m}^2, \\ h_a &= \sum_{m=-j}^{j-1} c_m v_{a,m} v_{a,m+1}. \end{aligned} \tag{A1}$$

Then

$$T_j = \frac{4}{d} \max_{v_a \geq 0,\ \sum_a v_{a,m}^2 = 1} \sum_a \frac{h_a^2}{n_a}. \tag{A2}$$

Indeed, for a complex rank-one vector w its trace contribution is $4|\langle w|J_+|w\rangle|^2/(d\|w\|^2)$. Replacing each coordinate by its modulus increases this quantity and preserves diagonal populations. Completeness is then restored exactly by the phase orbit

$$\begin{aligned} E_{a\ell} &= \frac{1}{L}|v_{a\ell}\rangle\langle v_{a\ell}|, \qquad v_{a\ell,m} = v_{a,m}e^{-im\phi_\ell}, \\ \phi_\ell &= \frac{2\pi\ell}{L}, \qquad L \geq \max(d, 3). \end{aligned} \tag{A3}$$

All nonzero frequencies $m - m'$ have absolute value below L, and their phase averages vanish. The first and second phase moments give zero score mean and an isotropic Fisher matrix. This argument also covers half-integer spins, since $m - m'$ is always an integer.

The maximum in (A2) is attained. To see this, put $p_{a,m} = v_{a,m}^2/n_a$ and $\mu_a = n_a/d$. These define a probability measure on the compact probability simplex with mean $(1/d, \ldots, 1/d)$. Its objective is the integral of the continuous function $4(\sum_m c_m\sqrt{p_m p_{m+1}})^2$. Weak compactness gives an optimizer. A supporting affine function for its concave envelope at the uniform vector, followed by Carathéodory's theorem on the contact set, reduces the support to at most d seeds. We will use attainment only when discussing strict finite-dimensional gaps.

Appendix B: A sum-of-squares lower bound

This appendix establishes the all-POVM lower bound used in Theorem 2 and defines the capped quantities entering the NH value in Section V. Index the d nodes by $i = 0, \ldots, d-1$ and the $d-1$ edges by $i = 0, \ldots, d-2$. Thus $c_i = \sqrt{(i+1)(d-1-i)}$. For $j > 0$, let τ_j be the unique positive solution of

$$\sum_{i=0}^{d-2} (c_i - \tau_j)_+ = \tau_j. \tag{B1}$$

Write

$$\begin{aligned} a_i &= \min(c_i, \tau_j), \qquad S_j = \sum_i c_i a_i, \\ D_j &= \sum_i c_i^2 - S_j, \qquad L_j = \frac{D_j}{d}. \end{aligned} \tag{B2}$$

Set all four quantities to zero when $j = 0$.

Proposition 4. *For every spin, $V_j \geq L_j$. Moreover,*

$$L_j = \frac{j}{2} + o(j), \qquad 0 \leq L_j \leq \frac{j}{2} + \frac{1}{4}. \tag{B3}$$

Proof. The left side of (B1) minus its right side is continuous and strictly decreasing, positive at zero and negative at $\max_i c_i$. Define

$$r_0 = 0, \qquad r_{i+1} - r_i = c_i/a_i - 1.$$

Equation (B1) gives $r_{d-1} = 1$ and $0 \leq r_i \leq 1$. Put $a_{-1} = a_{d-1} = 0$ and $\lambda_i = (1-r_{i-1})a_{i-1}^2 + r_{i+1}a_i^2$, omitting absent boundary terms. If K has off-diagonal entries $K_{i,i+1} = c_i$, direct expansion gives, for every real v and t,

$$\begin{aligned} &v^{\mathsf{T}}(\operatorname{diag}\lambda + t^2 I - tK)v \\ &\quad = \sum_{i=0}^{d-1} \big[(1-r_i)(tv_i - a_i v_{i+1})^2 \\ &\qquad + r_i(tv_i - a_{i-1}v_{i-1})^2\big] \geq 0. \end{aligned} \tag{B4}$$

Here absent vector components are zero, and $a_i(1 - r_i + r_{i+1}) = c_i$. Also $\sum_i \lambda_i = S_j$. For a nonzero seed take $t = h_a/n_a$ in (B4). Then $h_a^2/n_a \leq \sum_i \lambda_i v_{a,i}^2$. Summing over the complete seed family yields $T_j \leq 4S_j/d$. Since $\sum_i c_i^2 = \frac{2}{3}j(j+1)d$, Lemma 1 proves $V_j \geq D_j/d$. Thus (B4) is itself a certificate for the entire POVM optimization.

For the asymptotic statement let $J = j + 1/2 = d/2$. Then $c_i^2 = J^2 - (i+1-J)^2$ and $\tau_j < J$. The cap equation implies

$$\tau_j^2 \leq D_j = \sum_i c_i(c_i - \tau_j)_+ \leq J\tau_j \leq J^2. \tag{B5}$$

If $\tau_j \leq (1-\eta)J$ on an unbounded subsequence for some $\eta > 0$, a fixed positive fraction of the central edges satisfy $c_i \geq (1 - \eta/2)J$. Their contribution to $\sum_i (c_i - \tau_j)_+$ is of order J^2, contradicting (B1). Hence $\tau_j/J \to 1$ and $D_j/J^2 \to 1$. Dividing (B5) by $d = 2J$ proves both claims. □

Appendix C: Global Gaussian energy estimates

This appendix proves the Gaussian variance upper bound used in Theorem 2. The denominator in (15) couples all columns; a lower bound on h^2/s separates them into energies that can be estimated uniformly. Set

$$\begin{aligned} q(t) &= \sqrt{(J^2-t^2)_+}, \qquad B_m = \sqrt{J^2-m^2}, \\ k_m &= \frac{c_{m-1}+c_m}{2}, \end{aligned} \tag{C1}$$

where $c_{-j-1} = c_j = 0$. From $h^2/s \geq 2qh - q^2 s$,

$$\begin{aligned} W_j[b] &\leq U_j[b] \\ &:= \frac{1}{d}\int_{\mathbb{R}} \left[\sum_m (A_m + q^2)v_m^2 - 2qh_j\right] dt \\ &= \frac{1}{d}\left(\sum_m I_m + \sum_{m=-j}^{j-1} G_m + \sum_m R_m\right), \end{aligned} \tag{C2}$$

where every term is nonnegative:

$$\begin{aligned} I_m &= \int_{\mathbb{R}} (k_m - q(t))^2 v_m(t)^2 dt, \\ G_m &= c_m \int_{\mathbb{R}} q(t)(v_{m+1}(t) - v_m(t))^2 dt, \\ R_m &= \frac{(c_m - c_{m-1})^2}{4}. \end{aligned} \tag{C3}$$

To verify the second equality in (C2), use

$$\begin{aligned} h_j &= \sum_m k_m v_m^2 - \frac{1}{2}\sum_m c_m (v_{m+1} - v_m)^2, \\ A_m - k_m^2 &= R_m. \end{aligned}$$

Theorem 5. *For every fixed positive $b \in C^2([-1,1])$,*

$$\lim_{j\to\infty} \frac{U_j[b]}{j} = \frac{1}{2}\int_{-1}^{1} \left(x^2 b(x) + \frac{1}{4b(x)}\right) dx. \tag{C4}$$

For $b_\epsilon(x) = (2\sqrt{x^2+\epsilon^2})^{-1}$ and the explicit sequence (10),

$$\frac{U_j[b_{\epsilon_j}]}{j} \longrightarrow \frac{1}{2}, \qquad W_j[b_{\epsilon_j}] = \frac{j}{2} + o(j). \tag{C5}$$

1. Uniform control at the poles

All constants denoted C_b in this subsection depend only on a positive lower bound for b and an upper bound for its C^2 norm. For $r(s) = J - |s| \geq 1/2$,

$$\begin{aligned} Jr(s) \leq B(s)^2 &:= J^2 - s^2 \leq 2Jr(s), \\ S(s) &\asymp_b r(s), \qquad |S'(s)| \leq C_b, \\ |S''(s)| &\leq C_b/J. \end{aligned} \tag{C6}$$

The positive-part map is 1-Lipschitz. Therefore, for every real t, including $|t| \geq J$,

$$|q(t) - B_m| \leq \frac{|t^2 - m^2|}{B_m}. \tag{C7}$$

Under $v_m(t)^2 dt$, $t-m$ is normal with variance $S_m = S(m)$. Taking its second and fourth moments gives

$$\int (q - B_m)^2 v_m^2 dt \leq \frac{4m^2 S_m + 3S_m^2}{B_m^2} \leq C_b J. \tag{C8}$$

Since $c_m = q(m+1/2)$, the same inequality gives $|k_m - B_m|^2 \leq CJ/r(m)$. Consequently, for every column,

$$0 \leq I_m \leq C_b J. \tag{C9}$$

For the gradient term write $Y = (t-s)/\sqrt{S(s)}$. Direct differentiation yields

$$\partial_s v_s = v_s \left[\frac{Y}{2\sqrt{S}} + \frac{S'}{4S}(Y^2 - 1)\right]. \tag{C10}$$

The square-root Hölder inequality and (C6) imply

$$\begin{aligned} q(s + \sqrt{S}Y) &\leq B(s) + \sqrt{2J\sqrt{S}|Y| + SY^2} \\ &\leq C_b\sqrt{Jr(s)}(1 + |Y|). \end{aligned} \tag{C11}$$

This bound also includes all tails crossing the poles. Gaussian moments in (C10) now give

$$\begin{aligned} &\int q(t)(\partial_s v_s(t))^2 dt \\ &\quad \leq C_b\sqrt{Jr(s)}\left(\frac{1}{S(s)} + \frac{S'(s)^2}{S(s)^2}\right) \\ &\quad \leq C_b\sqrt{\frac{J}{r(s)}}. \end{aligned} \tag{C12}$$

For $s \in [m, m+1]$, $c_m \leq C\sqrt{Jr(s)}$. Integral Cauchy–Schwarz in s therefore yields

$$\begin{aligned} 0 \leq G_m &\leq c_m \int_m^{m+1} \int q(t)(\partial_s v_s(t))^2 dt\, ds \\ &\leq C_b J. \end{aligned} \tag{C13}$$

Finally $R_m \leq CJ/r(m)$, whence

$$\frac{1}{d}\sum_m R_m = O(\log(2+J)) = o(J). \tag{C14}$$

For fixed $0 < \delta < 1$, the columns with $|m| > (1-\delta)J$ and their incident edges thus contribute at most $C_b(\delta + J^{-1})$ to $U_j[b]/J$.

2. The interior limit

Fix $\delta > 0$ and restrict to $x_m = m/J$ with $|x_m| \le 1-\delta$. All errors below are uniform over this set; constants may depend on b and δ. Put $f(x) = \sqrt{1-x^2}$. Centered expansion gives $k_m - B_m = O_\delta(J^{-1})$. A global remainder estimate is

$$\begin{aligned}&\left|q(m+z) - B_m + \frac{m}{B_m}z\right| \\ &\quad\le \left(\frac{1}{2B_m} + \frac{4m^2}{B_m^3}\right)z^2 + \frac{z^4}{B_m^3}.\end{aligned} \tag{C15}$$

This follows from $|\sqrt{u_+} - B - (u-B^2)/(2B)| \le (u-B^2)^2/(2B^3)$, valid for every real u and $B>0$, by taking $u = J^2 - (m+z)^2$. Thus the estimate also covers the region where q vanishes. Set $z = \sqrt{Ja(x_m)}Y$. Its right-hand side is $O_{b,\delta}(Y^2 + J^{-1}Y^4)$. Squaring and using the Gaussian moments through order eight gives

$$\begin{aligned}\frac{I_m}{J} &= a(x_m)f'(x_m)^2 + O_{b,\delta}(J^{-1/2}) \\ &= x_m^2 b(x_m) + O_{b,\delta}(J^{-1/2}).\end{aligned} \tag{C16}$$

For the gradient term, use the weighted norm $\|w\|_q^2 = \int q(t)|w(t)|^2 dt$. In an interior unit interval, $S \asymp_{b,\delta} J$, $S' = O_b(1)$, and $S'' = O_b(J^{-1})$. Equation (C10) and one further derivative imply

$$\begin{aligned}\|\partial_s v_s\|_q &= O_{b,\delta}(1), \\ \|\partial_s^2 v_s\|_q &= O_{b,\delta}(J^{-1/2}).\end{aligned} \tag{C17}$$

Indeed, $\partial_s^2 v_s / v_s$ is a polynomial in Y of degree at most four, with coefficients $O_{b,\delta}(J^{-1})$, and $q \le J$. Two applications of the fundamental theorem of calculus and Minkowski's inequality give

$$\begin{aligned}&\|v_{m+1} - v_m - \partial_m v_m\|_q \\ &\quad\le \int_m^{m+1}\int_m^s \|\partial_u^2 v_u\|_q du\, ds \\ &\quad= O_{b,\delta}(J^{-1/2}).\end{aligned} \tag{C18}$$

Thus the squared norm of $v_{m+1} - v_m$ differs from that of $\partial_m v_m$ by $O_{b,\delta}(J^{-1/2})$. In (C10), the squared translation term contributes

$$\begin{aligned}&\frac{1}{4Ja(x_m)}\mathbb{E}\big[q(m+\sqrt{Ja(x_m)}Y)Y^2\big] \\ &\quad= \frac{f(x_m)}{4a(x_m)} + O_{b,\delta}(J^{-1/2}).\end{aligned}$$

The cross term containing S' is $O_{b,\delta}(J^{-1/2})$ and its square is $O_{b,\delta}(J^{-1})$. Since $c_m/J = f(x_m) + O_\delta(J^{-1})$, we obtain

$$\frac{G_m}{J} = \frac{1}{4b(x_m)} + O_{b,\delta}(J^{-1/2}). \tag{C19}$$

The Riemann sums in (C16) and (C19), divided by $d = 2J$, give the integral in (C4) over the interior. Equations (C9), (C13), and (C14) allow $\delta \downarrow 0$ after $j \to \infty$. This proves (C4), since $J/j \to 1$.

3. One explicit width sequence

Pointwise minimization of the integrand in (C4) suggests $b_*(x) = 1/(2|x|)$ away from the equator. Use instead

$$\begin{aligned}b_\epsilon(x) &= \frac{1}{2\sqrt{x^2+\epsilon^2}}, \\ a_\epsilon(x) &= \frac{1-x^2}{2\sqrt{x^2+\epsilon^2}}, \qquad 0 < \epsilon \le 1.\end{aligned} \tag{C20}$$

An exact integral evaluates the limiting functional:

$$\frac{1}{2}\int_{-1}^{1}\left(x^2 b_\epsilon + \frac{1}{4b_\epsilon}\right)dx = \frac{1}{2}\sqrt{1+\epsilon^2}. \tag{C21}$$

For fixed ϵ this already implies $\limsup_j V_j/j \le \sqrt{1+\epsilon^2}/2$. Together with Proposition 4, it proves (8). To specify a single attaining family, we retain the parameter dependence of the estimates above.

There is an absolute constant $C > 0$ such that whenever $0 < \delta < 1/2$, $J \ge \epsilon^{-1}$, and $J\delta \ge 1$,

$$\begin{aligned}&\left|\frac{U_j[b_\epsilon]}{J} - \frac{1}{2}\sqrt{1+\epsilon^2}\right| \\ &\quad\le C\epsilon^{-7}(\delta + J^{-1}) \\ &\qquad + C(\epsilon^{-2}\delta^{-6} + \epsilon^{-12}\delta^{-4})J^{-1/2} + C\frac{1+\log(2J)}{J}.\end{aligned} \tag{C22}$$

We give explicit bounds supporting these exponents. The formulas in (C20) give

$$\begin{aligned}&\frac{1}{2\sqrt{2}} \le b_\epsilon \le \frac{1}{2\epsilon}, \\ &\|a_\epsilon'\|_\infty \le C\epsilon^{-3}, \qquad \|a_\epsilon''\|_\infty \le C\epsilon^{-5}.\end{aligned} \tag{C23}$$

Equation (C8) then gives $I_m \le C\epsilon^{-2}J$. In (C11) one may take the constant $C\epsilon^{-1/2}$. Using $S \ge cr$ and $|S'| \le C\epsilon^{-3}$ in (C12) gives $G_m \le C\epsilon^{-7}J$. The deleted pole regions therefore produce the first term of (C22); the corresponding integral tails are at most $C\epsilon^{-1}\delta$.

In the interior, $S \ge cJ\delta$, $S \le CJ\epsilon^{-1}$, and S', JS'' have the polynomial bounds (C23). To check all derivative constants, write the bracket in (C10) as $g = AY + B(Y^2-1)$, where $A = (2\sqrt{S})^{-1}$ and $B = S'/(4S)$. Then $\partial_s^2 v_s / v_s = g^2 + \partial_s g$, and

$$\begin{aligned}\partial_s Y &= -S^{-1/2} - \frac{S'}{2S}Y, \qquad A' = -\frac{S'}{4S^{3/2}}, \\ B' &= \frac{S''}{4S} - \frac{S'^2}{4S^2}.\end{aligned}$$

The four types of coefficient in $\partial_s^2 v_s / v_s$ are S^{-1}, $S'/S^{3/2}$, S''/S, and S'^2/S^2. Hence, using $q \le J$ and fixed Gaussian moments,

$$\begin{aligned}\|\partial_s v_s\|_q &\le C\epsilon^{-3}\delta^{-1}, \\ \|\partial_s^2 v_s\|_q &\le C\epsilon^{-6}\delta^{-2}J^{-1/2}.\end{aligned} \tag{C24}$$

The finite-difference replacement therefore changes the squared norm by at most $C\epsilon^{-12}\delta^{-4}J^{-1/2}$. For the position term, use the global remainder (C15), $S \le CJ\epsilon^{-1}$, $B_m \ge J\sqrt{\delta}$, and $|k_m - B_m| \le CJ^{-1}\delta^{-3/2}$. Gaussian moments through order eight give the uniform bound

$$\left|I_m/J - x_m^2 b_\epsilon(x_m)\right| \le C\epsilon^{-2}\delta^{-6}J^{-1/2}. \tag{C25}$$

For the gradient expectation one can use the global bound (C7) directly:

$$\begin{aligned}&|q(m+\sqrt{S}Y) - B_m| \\ &\quad\le C\left[\sqrt{J}\epsilon^{-1/2}\delta^{-1/2}|Y| + \epsilon^{-1}\delta^{-1/2}Y^2\right].\end{aligned}$$

The errors in the squared translation term, the cross term with S', and its square are at most $C\epsilon^{-1}\delta^{-2}J^{-1/2}$, $C\epsilon^{-3}\delta^{-2}J^{-1/2}$, and $C\epsilon^{-6}\delta^{-2}J^{-1}$, respectively. Including (C24) and the change from c_m/J to $f(x_m)$,

$$|G_m/J - 1/(4b_\epsilon(x_m))| \le C\epsilon^{-12}\delta^{-4}J^{-1/2}. \tag{C26}$$

Finally the derivative of $x^2 b_\epsilon + 1/(4b_\epsilon)$ has absolute value at most $3/2$, uniformly in ϵ. This controls the Riemann-sum error. These are all terms in the preceding argument and prove (C22).

Now use (10) and, only in the proof, take $\delta_j = \epsilon_j^8$. The pole error is $O(1/\log J) + O((\log J)^7/J)$. The interior error is $O((\log J)^{50}/\sqrt{J})$. Every term tends to zero. Therefore $U_j[b_{\epsilon_j}]/j \to 1/2$. Since $L_j \le V_j \le W_j[b_{\epsilon_j}] \le U_j[b_{\epsilon_j}]$, (C5) follows. The cutoff δ_j has no role in the measurement; all columns and all real outcomes t remain present.

Appendix D: Schur averaging and completion of the main proof

This appendix completes Theorem 2 by averaging the spin bounds over the Schur probabilities. Write $\mathbb{E}_N$ for expectation under $P_{N,j}$. The following moment identities and remainder estimate control the sum as the number of sectors grows.

Lemma 6. *The Schur probabilities satisfy*

$$\begin{aligned}\mathbb{E}_N[j(j+1)] &= \frac{3N}{4}, \\ \frac{\mathbb{E}_N j}{\sqrt{N}} &\longrightarrow \sqrt{\frac{2}{\pi}}.\end{aligned} \tag{D1}$$

If a fixed sequence r_j, defined on all nonnegative integer and half-integer spins, has finite entries and $r_j = o(j)$, then $\mathbb{E}_N r_j = o(\sqrt{N})$.

Proof. The first identity is the trace of $(J_x^{\mathrm{tot}})^2 + (J_y^{\mathrm{tot}})^2 + (J_z^{\mathrm{tot}})^2$ in $I/2^N$. Let S_N be a sum of N independent symmetric signs. Summation by parts in $P_{N,j} = (2j+1)(q_j - q_{j+1})$, $q_j = 2^{-N}\binom{N}{N/2-j}$, gives the exact identity

$$\mathbb{E}_N j = \mathbb{E}|S_N| - \frac{1}{2} + \frac{1}{2}\Pr(S_N = 0). \tag{D2}$$

The binomial sums are

$$\mathbb{E}|S_{2n}| = 2n\frac{\binom{2n}{n}}{2^{2n}}, \qquad \mathbb{E}|S_{2n+1}| = (2n+1)\frac{\binom{2n}{n}}{2^{2n}}.$$

Stirling's formula proves (D1) for both parities. For any $\eta > 0$, choose a fixed K_η such that $|r_j| \le \eta j$ when $j \ge K_\eta$. If $M_\eta = \max_{j<K_\eta}|r_j|$, then

$$\frac{\mathbb{E}_N|r_j|}{\sqrt{N}} \le \eta\frac{\mathbb{E}_N j}{\sqrt{N}} + \frac{M_\eta}{\sqrt{N}}.$$

Letting first $N \to \infty$ and then $\eta \downarrow 0$ proves the claim. In particular, finitely many small-spin values cause no problem. □

By Theorem 5 and Lemma 6,

$$\mathbb{E}_N W_j^{\mathrm{G}} = \frac{1}{2}\sqrt{\frac{2}{\pi}}\sqrt{N} + o(\sqrt{N}). \tag{D3}$$

The same conclusion holds with W_j^{G} replaced by either V_j or L_j. For any of these three sequences Z_j,

$$\begin{aligned}\frac{4N}{2N - 4\mathbb{E}_N Z_j} &= 2 + \frac{4\mathbb{E}_N Z_j}{N} + o(N^{-1/2}) \\ &= 2 + \sqrt{\frac{8}{\pi}}N^{-1/2} + o(N^{-1/2}).\end{aligned} \tag{D4}$$

The lower bound holds for every POVM by Proposition 4, and the explicit family supplies the matching upper bound. This completes the proof of Theorem 2.

Appendix E: The NH formula and strict finite-copy separation

Here we supply the finite-dimensional facts used in Section V. The two-parameter NH bound can be expressed as a convex first-moment optimization, or equivalently as its usual semidefinite relaxation [1, 2]. Put $D = 2^N$ and $Z = X + iY$ for arbitrary Hermitian first moments. Eliminating the auxiliary operator blocks gives the objective

$$\begin{aligned}\mathcal{F}(Z) &= \min_{W \succeq Z^\dagger Z,\, W \succeq ZZ^\dagger} \operatorname{tr} W \\ &= \operatorname{tr} Z^\dagger Z + \max_{0 \preceq R \preceq I} \operatorname{tr} R(ZZ^\dagger - Z^\dagger Z).\end{aligned}$$

The equality follows by taking the positive spectral part of $ZZ^\dagger - Z^\dagger Z$. The normalized NH risk is (N/D) times the minimum of $\mathcal{F}$ under the first-moment constraints.

Use the Schur basis with each multiplicity label retained, and let R be diagonal with the entries r_i from Appendix B on each spin block. Then

$$\mathcal{F}(Z) \ge \sum_{\alpha,\beta}(1 + r_\alpha - r_\beta)|Z_{\alpha\beta}|^2.$$

Every coefficient is nonnegative. On a raising edge it is $1 - r_i + r_{i+1} = c_i/a_i$. The unbiasedness constraints imply

$\sum_{\text{raising edges}} c_i \operatorname{Re} Z_{i+1,i} = D$. Thus Cauchy–Schwarz on these edges gives $\mathcal{F}(Z) \geq D^2/S_{\text{tot}}$, where $S_{\text{tot}} = \sum_j \nu_j S_j$. This bound applies to all complex Z, including entries connecting different spin blocks.

To attain it, take adjacent raising matrices in each sector with real nonnegative entries $x_{j,i}$, identical on the multiplicity space. Set $w_j = P_{N,j}/d$. Their objective before multiplication by N is

$$\begin{aligned} &\sum_j w_j f_j(x_j), \\ f_j(x) &= \sum_i x_i^2 + \frac{1}{2}\sum_{i=0}^{d-1} |x_i^2 - x_{i-1}^2| \\ &= \sum_{i=0}^{d-1} \max(x_{i-1}^2, x_i^2). \end{aligned} \tag{E1}$$

where $x_{-1} = x_{d-1} = 0$. The zero means and cross derivatives vanish; the remaining unbiasedness constraint is $\sum_j w_j \sum_i c_i x_{j,i} = 1$. Expressions containing S_j^{-1} below include only $j > 0$ sectors; the $j = 0$ sector contributes zero. With r_i defined in Appendix B,

$$\begin{aligned} f_j(x) &\geq \sum_i (1 - r_i + r_{i+1}) x_i^2 \\ &= \sum_i \frac{c_i}{a_i} x_i^2 \geq \frac{(\sum_i c_i x_i)^2}{S_j}. \end{aligned} \tag{E2}$$

The first inequality supports each maximum by a convex combination of its two entries. At $x_i = a_i$, it is an equality: r_i selects the larger entry on each uncapped tail and mixes only two equal entries on the plateau. The second inequality is Cauchy–Schwarz and is also an equality there. A further weighted Cauchy–Schwarz inequality across sectors proves that the minimum is $1/\sum_j w_j S_j$, achieved by $x_{j,i} = a_{j,i}/\sum_k w_k S_k$. Multiplication by N proves (20).

Suppose now that the seed maximum (A2) attains the upper bound $4S_j/d$. Every seed then saturates the nonnegative terms of (B4) with $t = h/n \geq 0$. For $t > 0$, the kernel recurrence and connectedness force every coordinate of a nonzero seed to be positive. An edge with $c_i > \tau_j$ has $r_{i+1} > 0$ and $1 - r_i > 0$, so the two adjacent equations force $t = \tau_j$. All such seeds are therefore multiples of a single vector u. Its plateau coordinates can be set to one, and on the left tail

$$u_i = (a_i/\tau_j) u_{i+1}; \tag{E3}$$

the right tail is its reflection. For $t = 0$, the matrix in (B4) is diagonal and only its two endpoint entries can vanish; all interior λ_i are strictly positive.

If $c_1 < \tau_j$, the first two edges on the left are uncapped. Equation (E3) then forces every positive-t seed to satisfy $v_1 = (c_1/\tau_j) v_2$, while every zero-t seed vanishes at both nodes. Hence the sum of their effects obeys $M_{11} = (c_1/\tau_j)^2 M_{22}$, which is incompatible with $M = I$. Attainment of the seed maximum makes the resulting inequality strict.

To verify $c_1 < \tau_j$ for all $j \geq 5$, set $n = 2j \geq 10$. At $n = 10$ the six central edges give

$$\begin{aligned} \sum_i (c_i - c_1)_+ &= 2(\sqrt{24} + \sqrt{28} + \sqrt{30}) - 6\sqrt{18} \\ &> \sqrt{18} = c_1. \end{aligned}$$

For a fully rational check, the three ratios to $\sqrt{18}$ exceed respectively 23/20, 6/5, and 5/4; twice their sum minus six exceeds one. For each fixed $i \geq 2$,

$$\frac{c_i}{c_1} = \sqrt{\frac{(i+1)(n-i)}{2(n-1)}}$$

increases with n. The three edges $i = 2, 3, 4$ and their three reflections remain distinct for $n \geq 10$. Thus $\sum_i (c_i - c_1)_+ > c_1$ for every such n, and the monotonicity of (B1) implies $\tau_j > c_1$. For $N \geq 10$ the largest-spin sector $j = N/2$ has positive probability and a strict Fisher deficit, proving $C_N > C_N^{\text{NH}}$.

[1] M. Hayashi and Y. Ouyang, Tight Cramér–Rao type bounds for multiparameter quantum metrology through conic programming, Quantum **7**, 1094 (2023).

[2] J. Zhang and M. Hayashi, Schur–Weyl reduction of finite-copy precision bounds for multiparameter quantum estimation (2026), arXiv:2610.38299v1.

[3] Q. Y. He, S.-G. Peng, P. D. Drummond, and M. D. Reid, Planar quantum squeezing and atom interferometry, Phys. Rev. A **84**, 022107 (2011).

[4] L. Dammeier, R. Schwonnek, and R. F. Werner, Uncertainty relations for angular momentum, New J. Phys. **17**, 093046 (2015).

[5] G. Tóth and F. Fröwis, Uncertainty relations with the variance and the quantum Fisher information based on convex decompositions of density matrices, Phys. Rev. Research **4**, 013075 (2022).